

Name: _____ Date: _____

Period: _____

1. Let $f(x) = x^2 - 3$. Evaluate each expression.

(a) $f(0) = -3$

(b) $f(-2) = 1$

(c) $f(x+1) = (x+1)^2 - 3 = x^2 + 2x - 2$

2. The table below gives values of a function f .

x	-2	-1	0	1	2
$f(x)$	5	2	1	2	5

State the domain and range of f using the values in the table.

Domain: $\{-2, -1, 0, 1, 2\}$ Range: $\{1, 2, 5\}$

3. The graph of $p(x)$ is a polynomial with the following features: it passes through $(-2, 0)$, $(1, 0)$, and $(0, -2)$, and its end behavior is $p(x) \rightarrow +\infty$ as $x \rightarrow +\infty$ and $p(x) \rightarrow -\infty$ as $x \rightarrow -\infty$.

Circle the best description of the degree and leading coefficient of $p(x)$.

A. Even degree, positive leading coefficient B. **Odd degree, positive leading coefficient ← correct**

C. Even degree, negative leading coefficient D. Odd degree, negative leading coefficient

(End behavior $-\infty \rightarrow +\infty$ matches odd degree with positive leading coefficient.)